ON A GENUS ZERO CURVE EVERY DEGREE ZERO INVERTIBLE MODULE IS TRIVIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the Riemann-Roch input is Lemma lemma-rr (in-source, navigation stub); the trivialization step is the standard Varieties criterion (external tooling, stub).

Editorial note. The label lemma-genus-zero-pic names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Riemann-Roch. Same unit, Lemma lemma-rr. The proof applies the Riemann-Roch inequality of Lemma lemma-rr to the invertible module L.
- Trivialization criterion. Varieties, Lemma varieties-lemma-check-invertible-sheaf-trivial (standard tooling). Final step (source lines 2014-2016): an invertible sheaf with a nonzero global section is trivial, by the Varieties criterion.

Source reference. Original reference 0BS6: sources/stacks-curves/pinned-original-source.tex, source lines 1218–1218. Prerequisite pointer invoked at proof line(s) 2013; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0B40: sources/stacks-chow/tags.txt, source lines 13983–13983. Prerequisite pointer invoked at proof line(s) 2015; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a proper curve over a field k with $H^0(X, \mathcal{O}_X) = k$. If X has genus 0, then every invertible $\mathcal{O}_X$-module $\mathcal{L}$ of degree 0 is trivial.*

Proof. Namely, we have $\dim_k H^0(X, \mathcal{L}) \geq 0 + 1 - 0 = 1$ by Riemann-Roch (Lemma 0BS6), hence $\mathcal{L}$ has a nonzero section, hence $\mathcal{L} \cong \mathcal{O}_X$ by Varieties, Lemma 0B40. □

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.